

Shear Amplitude Calculation in Abaqus Simulation

Parameter Definitions

- μ : Friction coefficient, unitless ($\mu = 0.70$)
- σ_n : Normal stress, in MPa ($\sigma_n = 10$ MPa)
- Y : Yield strength of PMMA, in MPa ($Y = 3000$ MPa)
- A : Contact area between spring and plate, $A = \text{SPR_W} \times \text{SPR_D}$, in mm^2
- h : Height of the spring (SPR_H), in mm
- D : Depth of contact interface (DEPTH), in mm
- L : Rupture length along contact interface (RUPTURE_POSITION), in mm

Step 1: Frictional Resistance Force

$$F_{\text{resist}} = 2\mu \cdot \sigma_n \cdot D \cdot L \quad (1)$$

Unit Check:

$$\begin{aligned} [\text{MPa}] \cdot [\text{mm}] \cdot [\text{mm}] &= \frac{\text{N}}{\text{mm}^2} \cdot \text{mm} \cdot \text{mm} \\ &= \text{N} \end{aligned}$$

So, F_{resist} has unit of N.

Step 2: Shear Amplitude

The shear amplitude is computed from:

$$\delta = \frac{F_{\text{resist}} \cdot h}{A \cdot Y} \quad (2)$$

Unit Check:

$$\begin{aligned} \frac{[\text{N}] \cdot [\text{mm}]}{[\text{mm}^2] \cdot [\text{MPa}]} &= \frac{\text{N} \cdot \text{mm}}{\text{mm}^2 \cdot \frac{\text{N}}{\text{mm}^2}} \\ &= \text{mm} \end{aligned}$$

Thus, δ (shear amplitude) has unit of mm, which is physically consistent as a displacement.

Alternative Expression (Code Form)

In Python-style code:

```
RESISTANCE = 2 * FRICTION_COEFFICIENT * DEPTH * RUPTURE_POSITION * NORMAL_STRESS  
SHEAR_AMPLITUDE = RESISTANCE * SPR_H / ( SPR_W * SPR_D * Y_PMMA )
```

Or, more explicitly:

```
CONTACT_AREA = SPR_W * SPR_D  
SPRING_STIFFNESS = Y_PMMA * CONTACT_AREA / SPR_H  
SHEAR_AMPLITUDE = RESISTANCE / SPRING_STIFFNESS
```